Lecture 7

EE3010: Electrical Devices and Machines

School of Electrical and Electronic Engineering

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Learning Objectives

By the end of this lecture, you should be able to:

- Describe the problems associated with the commutation process in real DC machines.
- Explain the problems caused by armature reaction and the inductive voltage kick.
- Identify the solutions that may be used to overcome the problems associated with the commutation process.
- Describe the power flows and the power flow diagram of a DC machine.

Problems with Commutation in Real Machines

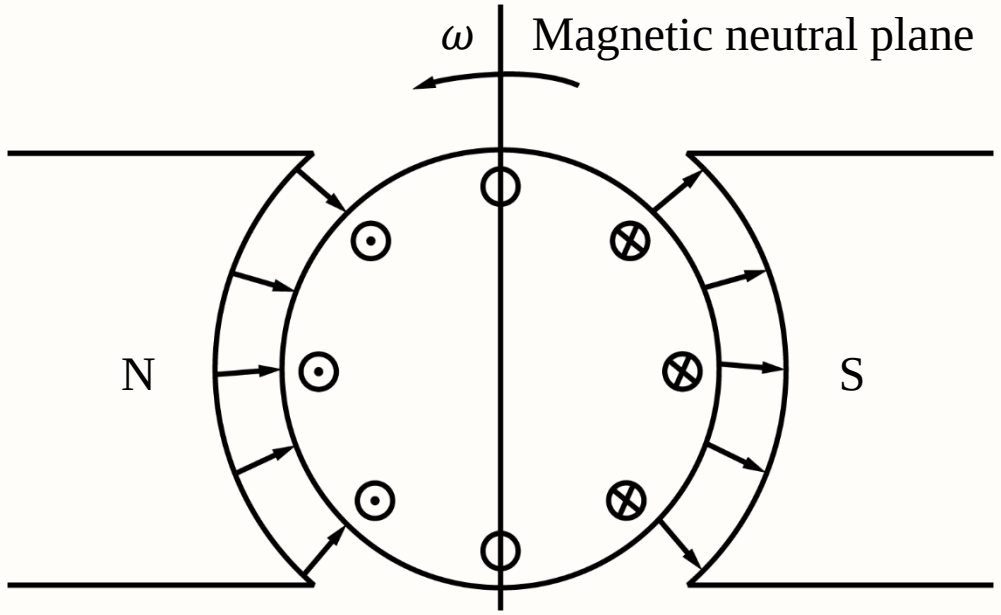
- Two major effects occur in the real world to disturb the commutation process:
  - Armature reaction
  - The $L\frac{di}{dt}$ voltages

Armature Reaction

- The current flowing in the armature windings will produce a magnetic field (armature mmf) of its own, which will distort the original magnetic field from the machine's poles. The effect produced by the armature mmf is called armature reaction.
- Problems caused by armature reaction:
  - Magnetic neutral plane shift
  - Flux weakening effect

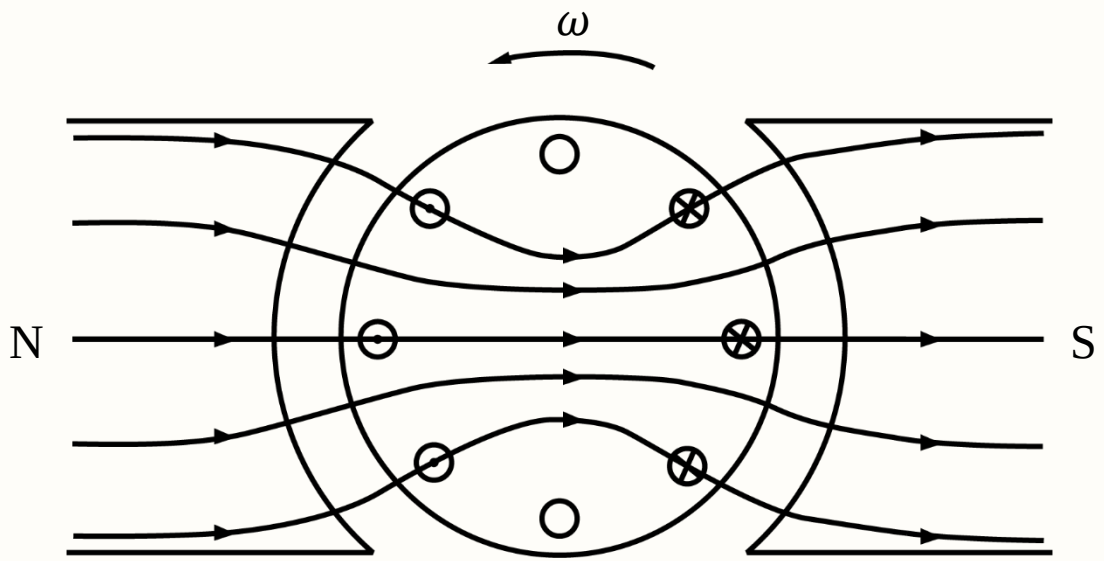
Magnetic Neutral Plane Shift

- The magnetic neutral plane is the position where the armature windings are moving parallel to the magnetic flux lines. They do not cut any lines of flux and there are no induced voltages.

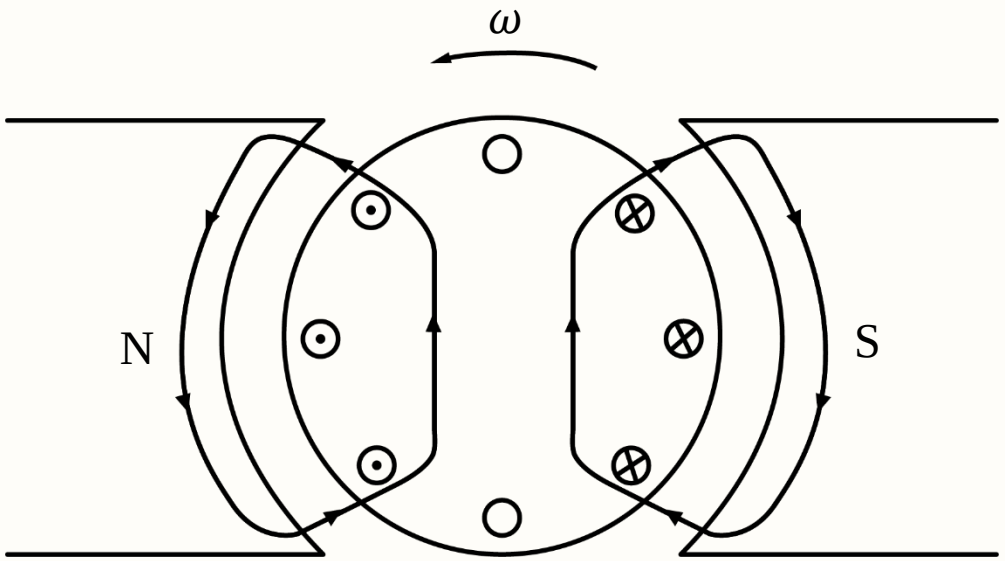


Initially, the pole flux is uniformly distributed in a DC generator, and the magnetic neutral plane is vertical.

Development of Armature Reaction in a DC Generator

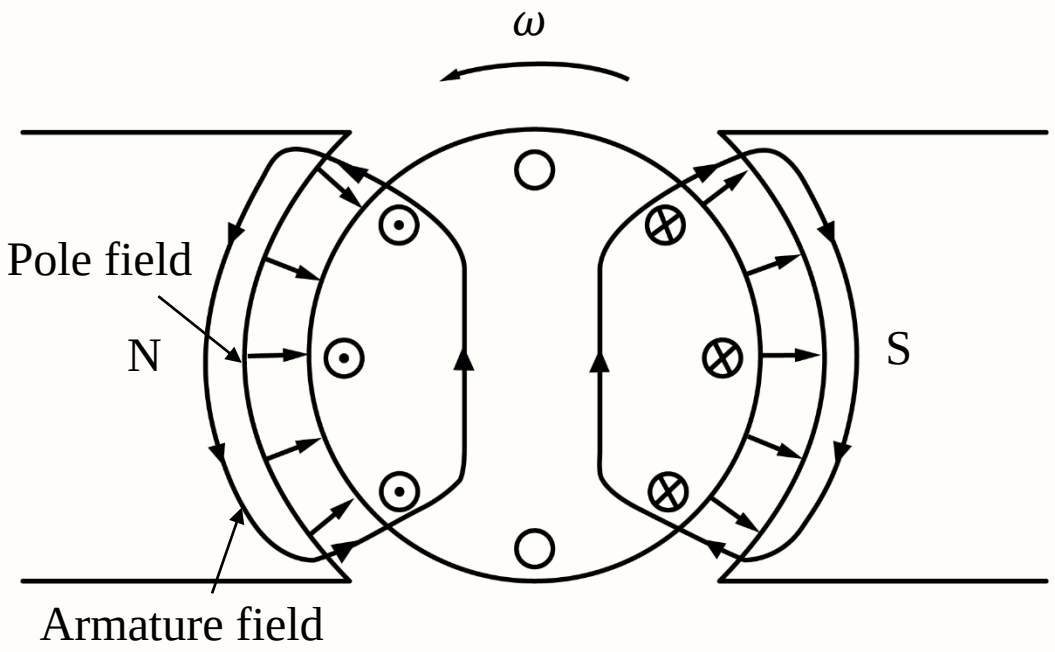


Effect of air gap on the pole flux distribution.

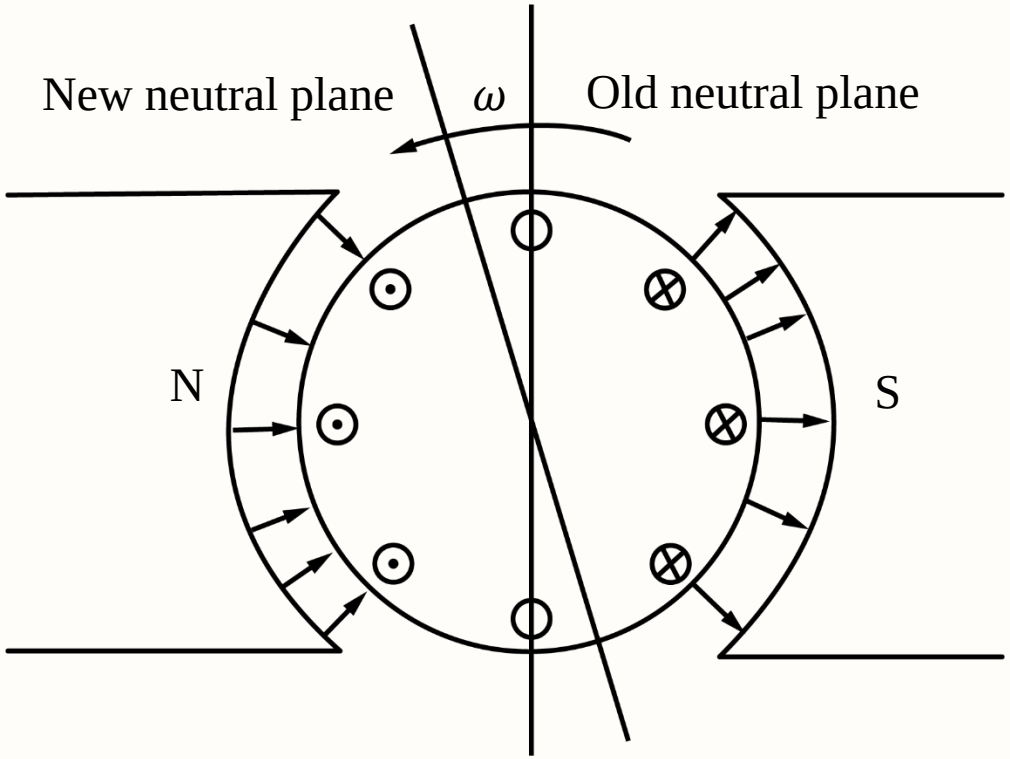


Magnetic field produced by currents flowing in armature conductors.

Development of Armature Reaction in a DC Generator



Rotor and pole fluxes adding and subtracting.



Resulting flux under the poles.
Neutral plane has shifted.

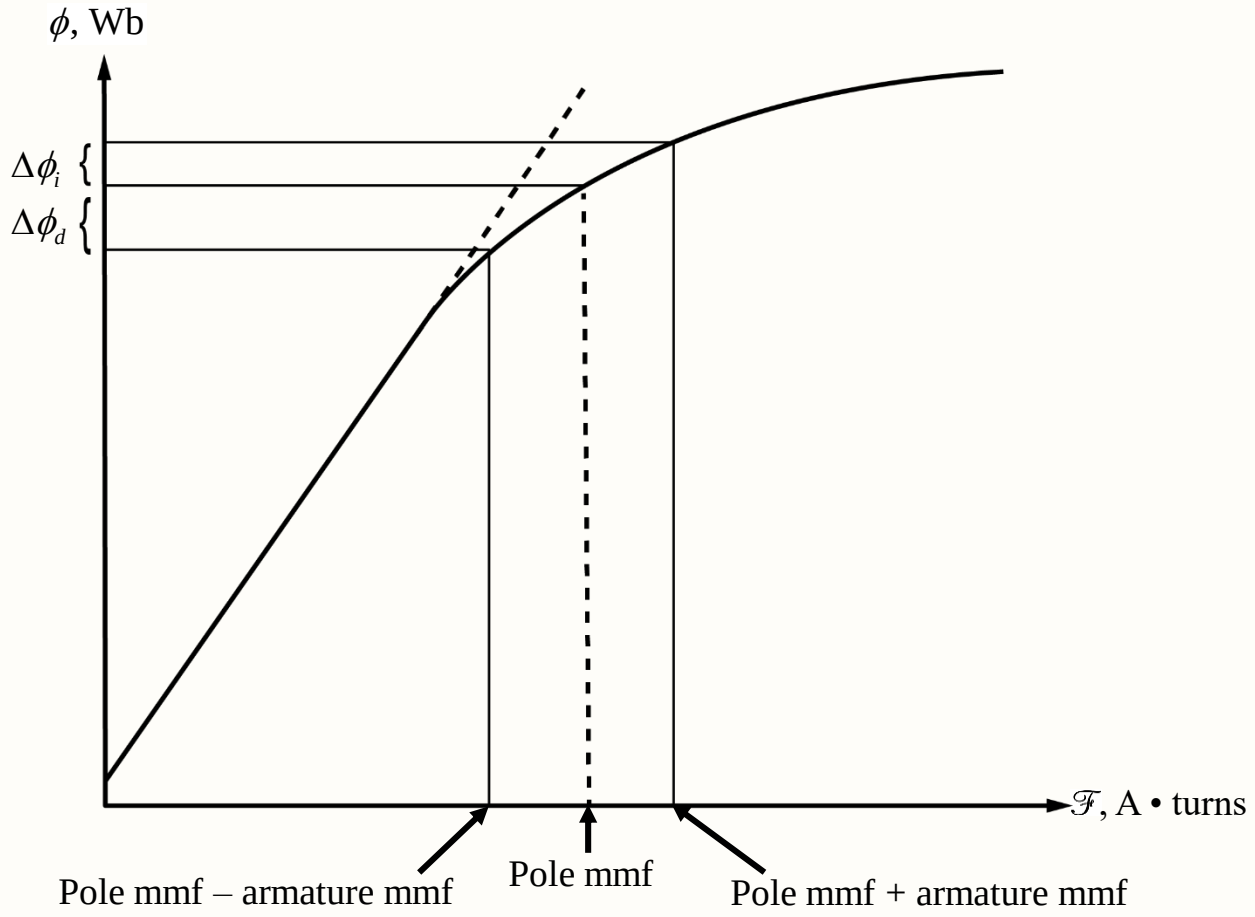
Magnetic Neutral Plane Shift

- The process of commutation requires the brushes shorting out the commutator segments just at the moment when the voltage across the segments is equal to zero.
- When the neutral plane shifts, the brushes short out commutator segments with a finite voltage across them. The result is a current flow circulating between the shorted segments.
- Severe arcing and sparking at the brushes may occur.

Flux Weakening Effect

- Most machines operate at flux densities near the saturation point.
  - At locations where rotor and pole mmfs add, only a small increase in flux occurs.
  - At locations where rotor mmf subtracts from the pole mmf, there is a larger decrease in flux.
- As a result, the total average flux under the entire pole face is decreased.

Flux Weakening Effect



$\Delta\phi_i \equiv$ flux increase under reinforced sections of poles

$\Delta\phi_d \equiv$ flux decrease under subtracting sections of poles

Flux Weakening Effect

- In generators, flux weakening reduces the voltage supplied by the generator for any given load.
- In motors, when flux is decreased, the speed of the motor increases.

$L\frac{di}{dt}$ Voltages

- They occur in commutator segments being shorted out by the brushes, sometimes called inductive kick.
- When a commutator segment is shorted out, the current flow through that commutator segment must reverse.
- The rate of change in current with respect to time in the shorted loop can be very high and a very significant inductive voltage kick will be induced in the shorted commutator segment.
- This high voltage naturally causes sparking at the brushes of the machine.

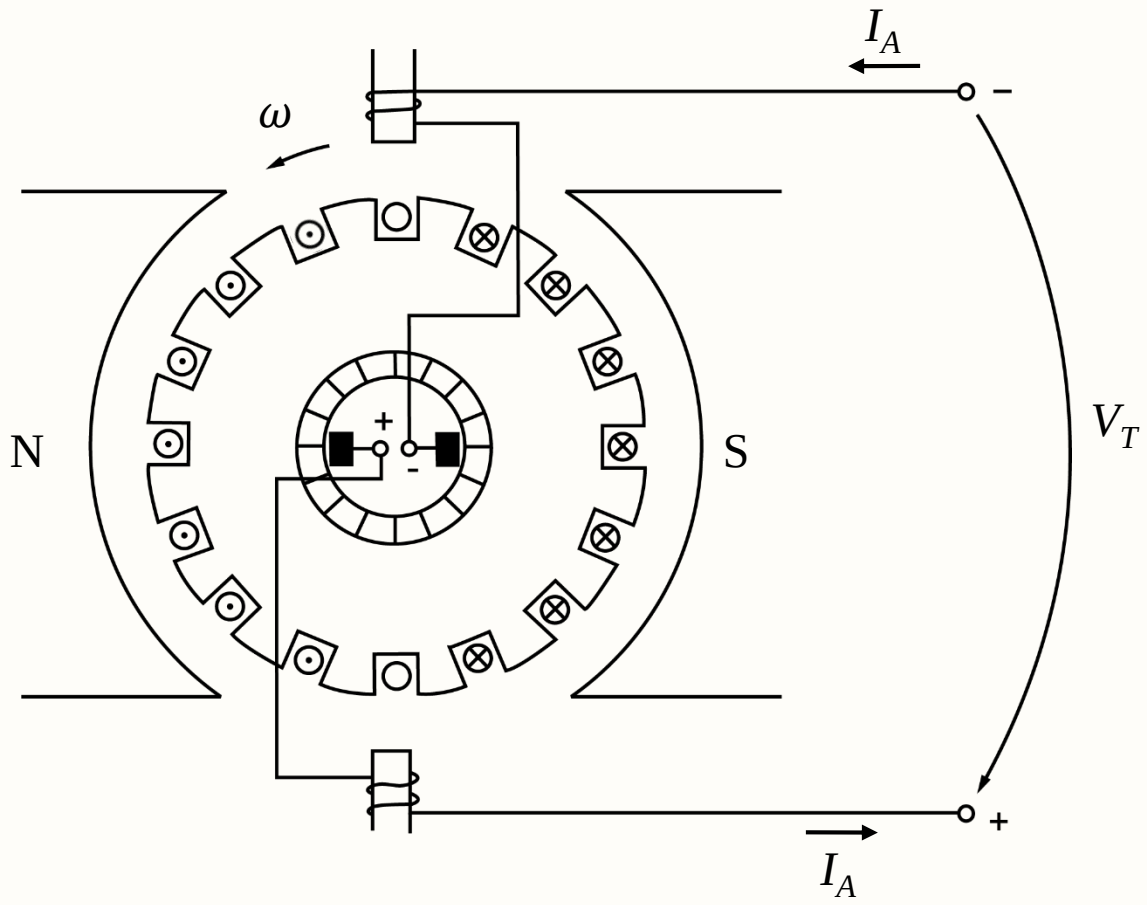
Solutions to the Problems with Commutation

- Brush shifting
  - Shifting the brushes to the new neutral plane to reduce sparking.
  - But if the load fluctuates, the armature mmfs fluctuate, and so the neutral plane shifts back and forth between no load and load conditions.
  - Someone has to adjust the brushes every time the load on the machine changes. This is not practical.

Solutions to the Problems with Commutation

- Commutating poles or interpoles
  - These small poles carry windings that are connected in series with the armature and are located directly over the conductors being commutated.
  - The number of turns on the windings is such that the interpoles developed an mmf that is equal and opposite to the armature mmf.
  - As the load current varies, the two mmfs rise and fall together, exactly bucking each other at all times. The voltage in the coils undergoing commutation can be exactly cancelled, leading to no sparking at the brushes.

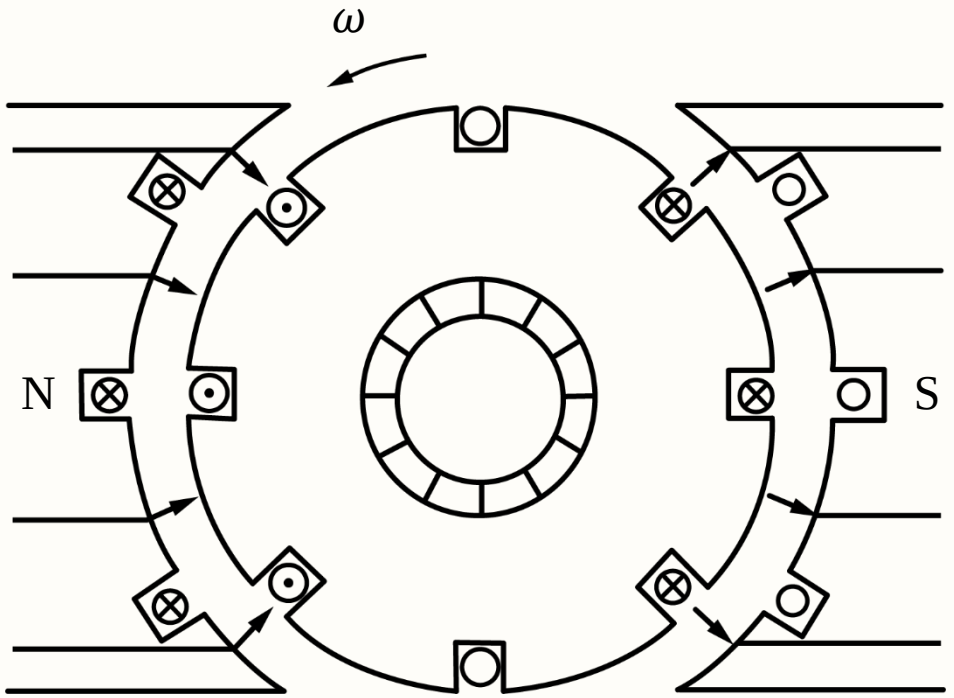
Solutions to the Problems with Commutation



A DC machine with interpoles, placed midway between the main poles.

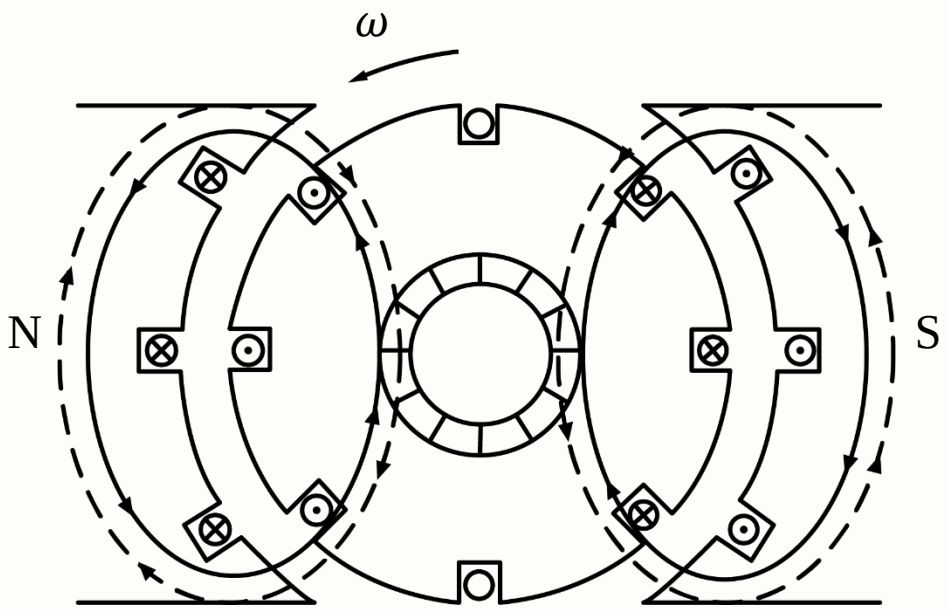
Solutions to the Problems with Commutation

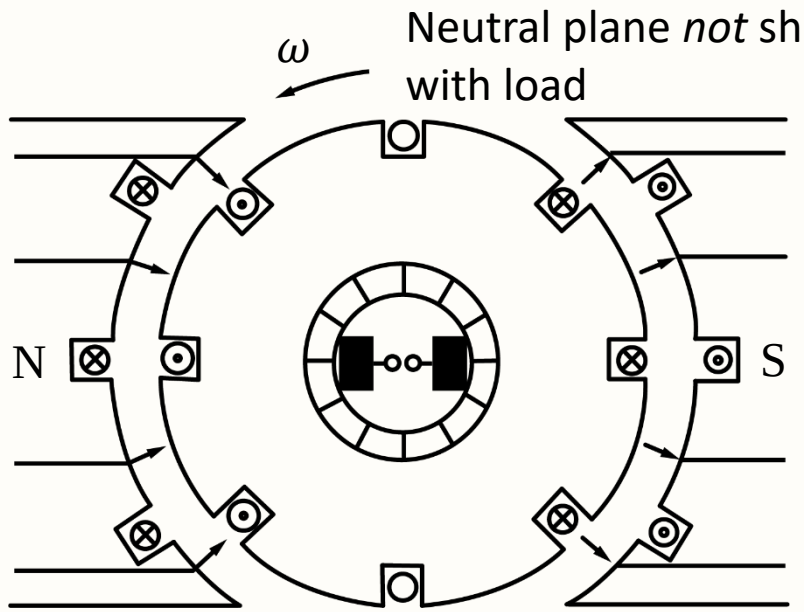
- Compensating windings
  - The windings are distributed in slots carved in the faces of the poles parallel to the rotor conductors and connected in series with rotor windings.



The pole flux in the machine.

Solutions to the Problems with Commutation





Rotor (armature) flux Flux from compensating windings

Fluxes from armature and compensating windings. They are equal and opposite and cancel each other out.

The net flux in the machine, which is just the original pole flux.

Power Flow and Losses in DC Machines

- DC generators take in mechanical power and produce electric power.
- DC motors take in electric power and produce mechanical power.
- Efficiency of a DC machine is defined as $\eta = \frac{P_{out}}{P_{in}} \times 100\% = \frac{P_{in} - P_{losses}}{P_{in}} \times 100\%$
- Losses in a DC Machine:
  a) Electrical or copper losses
  b) Brush losses
  c) Core losses
  d) Mechanical losses
  e) Stray load losses

Power Flow and Losses in DC Machines

a) Electrical or copper losses

- Losses that occur in the armature (rotor) and field (stator) windings of machine:

Armature copper loss: $P_A = I_A^2 R_A$

Field copper loss: $P_F = I_F^2 R_F$

Power Flow and Losses in DC Machines

b) Brush losses

- The power lost across the contact potential at the brushes of the machine. Relatively small.

$$P_{BD} = V_{BD} I_A$$

V_{BD} = brush voltage drop, usually assumed to be 2 V for graphite brushes

I_A = armature current

c) Core losses

- The hysteresis losses and eddy current losses occurring in the metal.

Power Flow and Losses in DC Machines

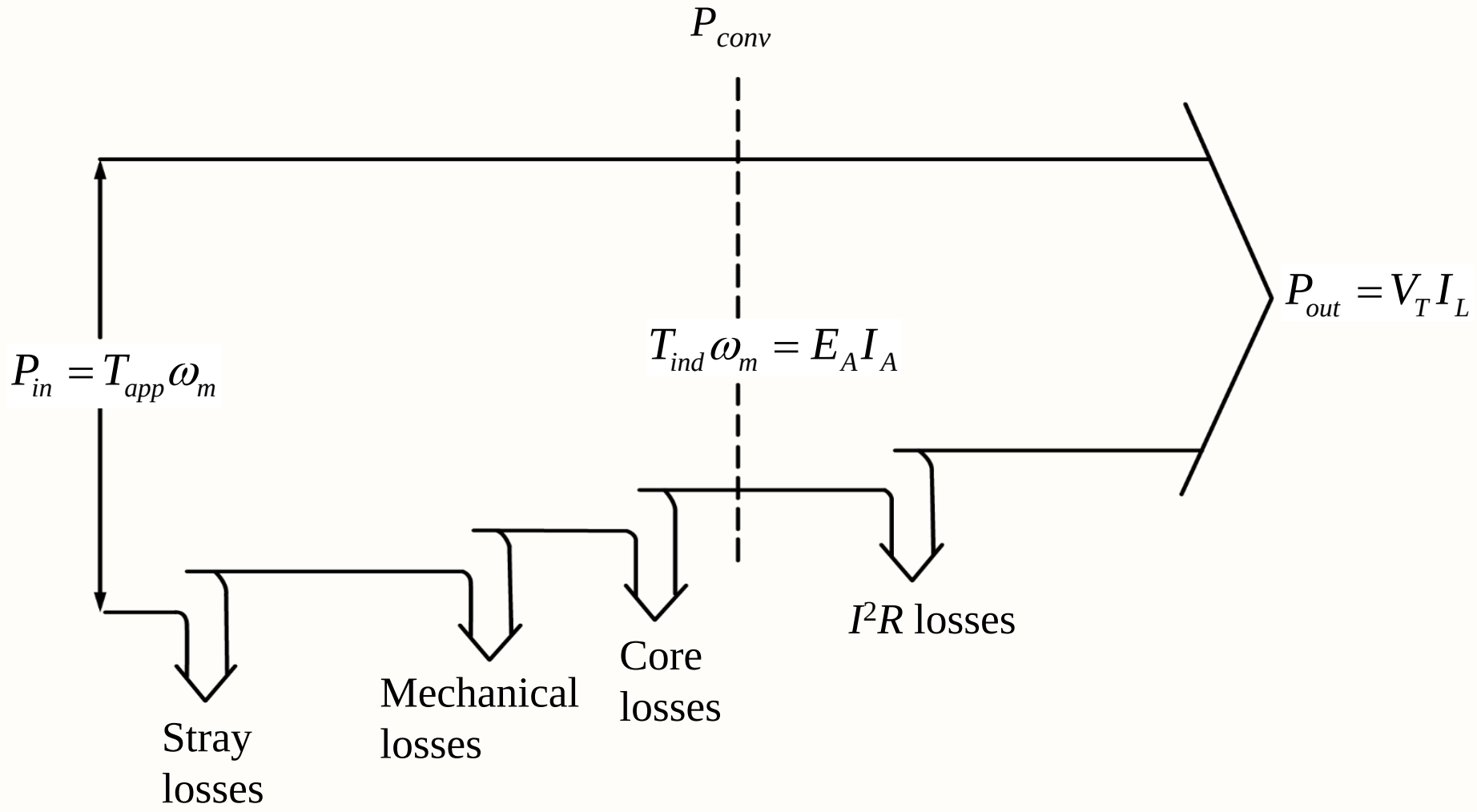
d) Mechanical losses

- Friction loss: caused by friction of the bearings in the machine and friction between brushes and commutator.
- Windage loss: caused by friction between the moving parts of the machine and the air inside the motor's casing.

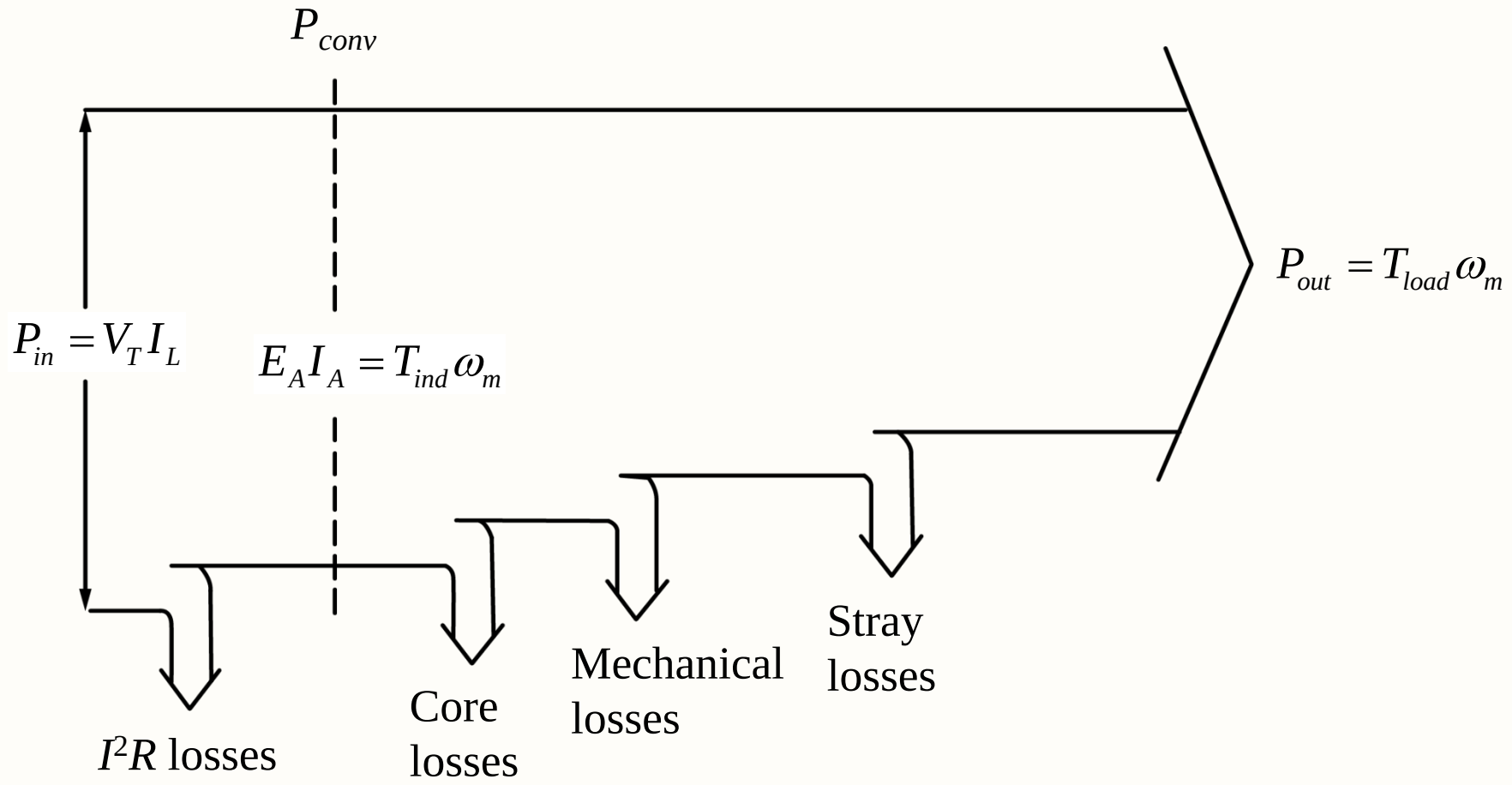
e) Stray (or miscellaneous) losses

- Any losses that cannot be placed in the above categories, e.g., losses in coils undergoing commutation. Approx 1% of full load (IEEE-Std 113-1985). In this course, if it is not mentioned, will assumed to be neglected.

Power Flow Diagram for DC Generator



Power Flow Diagram for DC Motor

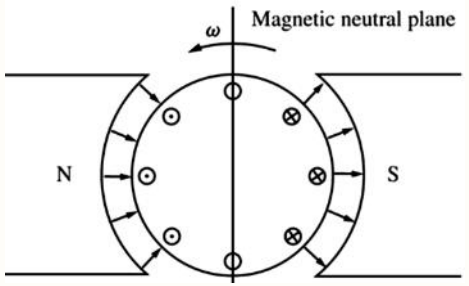
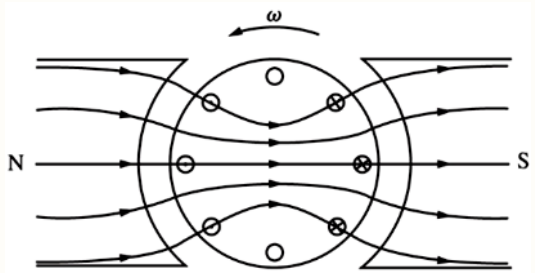
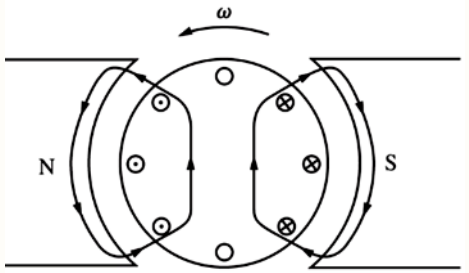


Summary

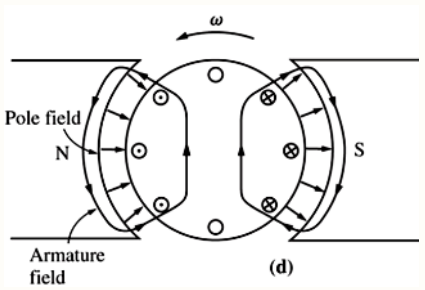
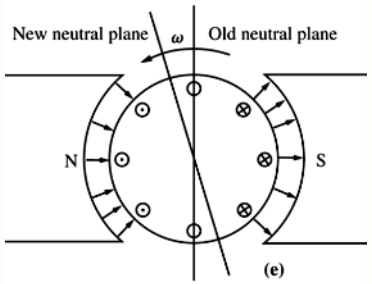
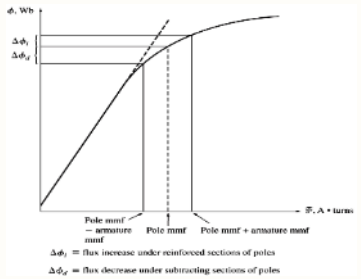
In this lecture, you have learnt:

- The problems caused by armature reaction and the inductive voltage kick in real DC machines.
- The solutions to the problems with commutation, such as utilising brush shifting, interpoles and compensating windings.
- The power flow diagram and losses of a DC machine.

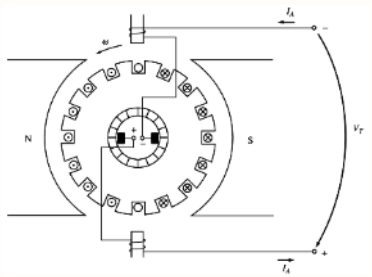
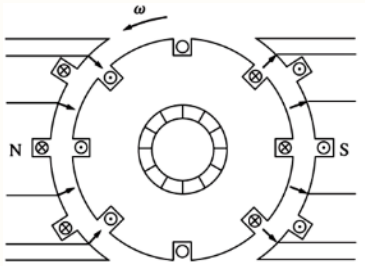
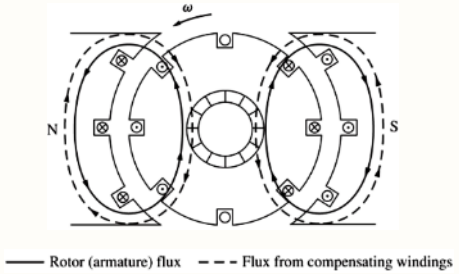
References

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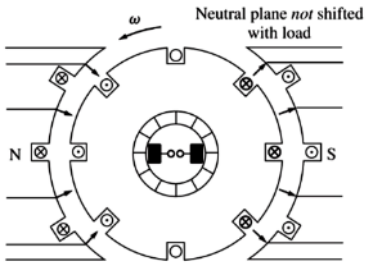
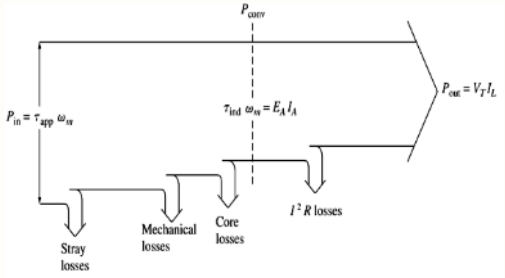
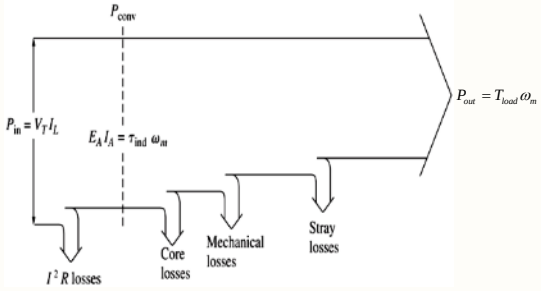
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